

LANDER NEURAL NETWORK PROJECT

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DATA OVERVIEW

- Features (Inputs): $X_{\text{dist}}, Y_{\text{dist}}$
- Labels (Outputs): $X_{\text{vel}}, Y_{\text{vel}}$
- Size: 70,000 records

PARTITIONING DATA

- **Train/Validation/Test split:**
70% Train, 15% Validation, 15% Test
- **Shuffling:** Randomized before splitting
- **Scaling:** Min-max normalization applied using training data min/max



ARCHITECTURE

Architecture/Topology:

- Input layer: 2 neurons
- Hidden layer: 7 neurons
- Output layer: 2 neurons
- Activation: Sigmoid

Loss Function: Mean Squared Error (MSE)

Hyperparameters:

- Learning rate: 0.5
- Momentum: 0.5
- Epochs: 100

Scaling:

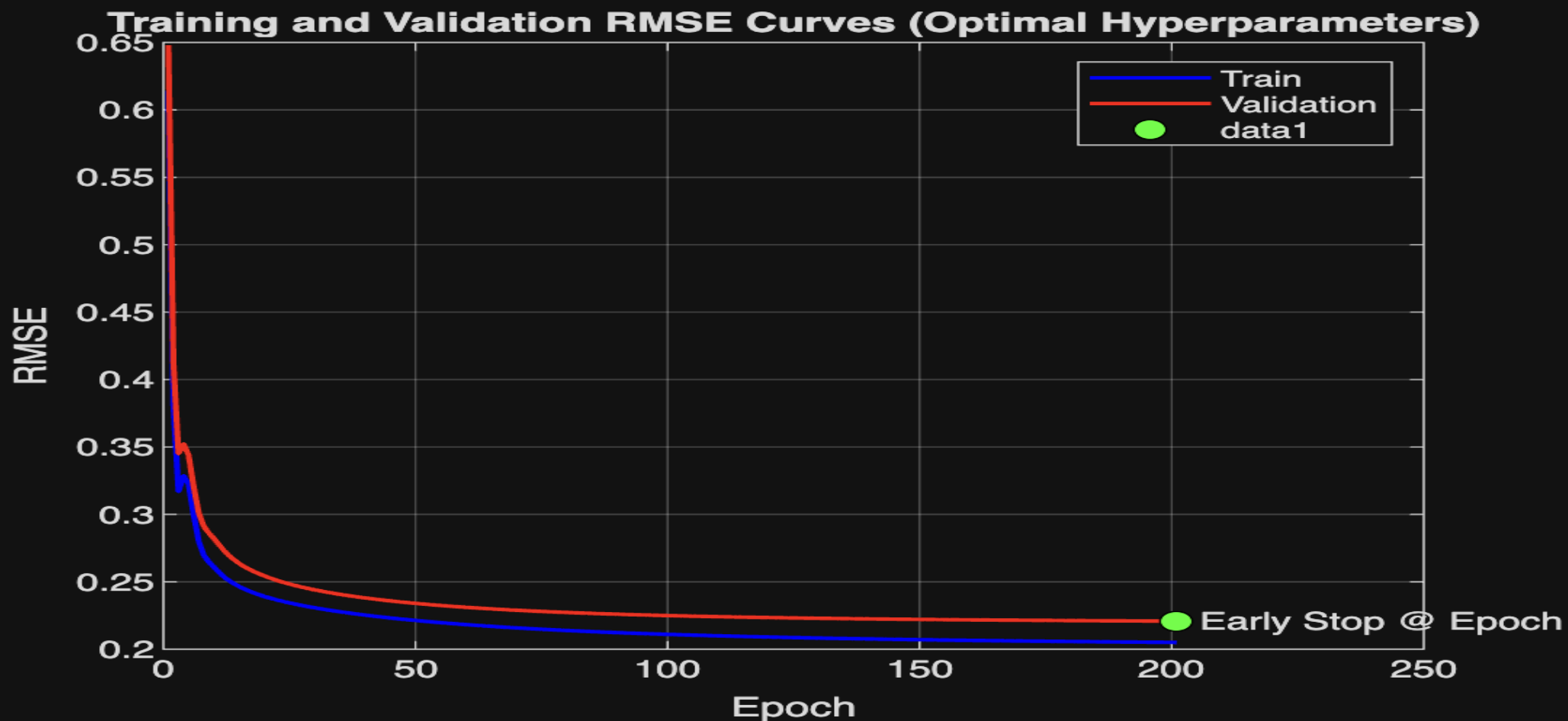
- MinMax Scaler

Best Hyperparameters:

hiddenNeurons: 7
lr: 0.5000
momentum: 0.5000
minGrad: 1.0000e-05
epochs: 200

Best Validation RMSE: 0.22083

Test RMSE: 0.20729



THANK YOU

Asha Deen